

Trip Analysis – Summary of Findings and Recommendations

Findings

The analysis was conducted on **200 ride records** to understand the relationships between trip distance, fare amount, ride category, surge pricing, customer ratings, and ride timing.

1. **Trip Distance and Fare:** Pearson correlation was **0.017** with a p-value of **0.813**, indicating no statistically significant relationship between trip distance and fare amount. Therefore, distance alone does not explain fare variation in this dataset.
2. **Premium vs Economy Rides:** The average fare for Premium rides was approximately **₹252.46**, compared with **₹257.75** for Economy rides. The t-test produced a p-value of **0.685**, showing that the difference is not statistically significant. ANOVA across Economy, Premium, and Shared categories also showed no significant difference (**p = 0.397**).
3. **Surge Pricing and Customer Ratings:** Spearman correlation was approximately **-0.047**, with a p-value of **0.508**. This indicates a very weak and statistically insignificant relationship between surge pricing and customer ratings.
4. **Peak vs Non-Peak Rides:** The comparison of fare amounts between Peak and Non-Peak rides produced a p-value of approximately **0.552**. Thus, there is insufficient statistical evidence that Peak rides generate significantly different fare revenue.
5. **Fare Anomalies:** The IQR method identified **one potential fare outlier**. However, no observation exceeded the ± 3 standard-deviation z-score threshold. The potential outlier should therefore be investigated rather than automatically removed.
6. **Ride Demand:** Economy rides represented approximately **41%** of the sample, Shared rides **30%**, and Premium rides **29%**. These probabilities provide a useful baseline for understanding ride-category demand.

Recommendations

- **Do not rely on trip distance alone for pricing.** Since distance has almost no correlation with fare in this sample, other factors should be considered.
- **Review the pricing structure** for Premium and Economy services, but avoid making major changes solely from this dataset because the observed differences are not statistically significant.
- **Monitor surge pricing and customer satisfaction.** Although the relationship with ratings is weak, surge pricing should be monitored continuously to identify potential customer-experience issues.
- **Investigate the identified fare anomaly** to determine whether it represents a genuine unusual ride or a data/pricing error.
- **Use demand probabilities for operational planning.** Economy rides have the highest observed demand, so vehicle availability and driver allocation can be planned accordingly.
- **Collect additional operational variables**, such as pickup/drop-off location, waiting time, driver availability, traffic conditions, vehicle type, booking time, and demand level.
- **Use advanced statistical models** such as multiple regression, hypothesis testing, confidence intervals, bootstrapping, and A/B testing before implementing significant pricing or service changes.

Overall Conclusion

The analysis indicates that the available variables have **weak statistical relationships with fare and customer ratings**. None of the major tested relationships was statistically significant at the 5% level. Therefore, the dataset is useful for establishing a baseline, but **additional operational data is required to make reliable pricing and service decisions**.